

Department of Artificial Intelligence & Data Science

School of Computing, FAST-NUCES

Course Title: Natural Language Processing (NLP)

Class: MS-AI/DS

Instructor: Dr. Zohair Ahmed

Total Marks: 10

Due Date: Next class. No extension. Late submissions will not be accepted.

- Viva or quiz may be conducted in the next class based on the assignment.
- Students must prepare the concepts.
- Blind copy-paste is not acceptable.

Task 1 Code

Instructions to Students

- **Setup and warm-up:**
 - Create Colab notebooks / environment.
 - Install `torch`, `transformers`, `datasets`, `scikit-learn`, `torchaudio`.
- **Dataset exploration:**
 - **Load** and inspect samples from IMDB, CoNLL-2003, SQuAD, and a small speech dataset (e.g., subset of LibriSpeech or Common Voice transcripts).
 - **Print statistics:** label distribution, sentence lengths, vocabulary size (approximate).